

REMOTE SENSING IMAGE ACQUISITION, ANALYSIS AND APPLICATIONS

Module Two

Computer-based interpretation—fundamentals of machine learning

SOLUTIONS TO END-OF-LECTURE QUIZZES

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Lecture 1. Fundamentals of image analysis and machine learning

- When would classification be used in remote sensing as an analysis tool?
When estimates of the areas of cover types is of interest and when maps of cover types are needed.
- In which of the following situations would it be useful to make decisions about the correct label for a pixel by being sensitive to the surrounding pixels:
 - In agriculture
 - In urban studies
 - For data with very high spatial resolution
 - For data with very low spatial resolution

Certainly, for agriculture where the field sizes are usually very large compared with the spatial resolution (pixel size) of the sensor. Usually less likely for urban regions unless very high spatial resolutions are being used. In general, spatial context is more important when we have image data with very high spatial resolution.

- Because of the radiometric resolution of the sensors used in remote sensing, the spectral space can be considered to consist of a set of discrete cells. For 8 bit data, how many cells are there if the number of bands is 4, 12, 220? For a 1000x1000 pixel image what can you say about the sparsity of the space in each case?
For 8 bit data there will be 256 levels of brightness in each band from 0 to 255. The number of discrete cells in the spectral space for that data is in the second column of the table.

bands	Number of cells	Cells per pixel for 1000x1000 pixel image
4	$256^4 = \sim 4.3 \times 10^9$	$\sim 4.3 \times 10^3$
12	$256^{12} = \sim 7.9 \times 10^{28}$	$\sim 7.9 \times 10^{22}$
220	$256^{220} = \text{huge!}$	A very, very, very large number

The last column shows that the data space is likely to be very sparse as the number of bands grows. Of course, there is an assumption here, in that we think of the pixels being spread uniformly across the multi-dimensional space, which they won't be. They will tend to cluster in ranges of brightness; nevertheless, those larger data spaces are very dilute.

Lecture 2. The maximum likelihood classifier

- Why do you think the normal distribution is usually chosen to represent clusters of pixels in spectral space?
Because its mathematical properties are well-known and it is easy to work with.
- Would there be value in using more than one multivariate normal distribution per class?

Yes indeed, because the data spread in a given class is likely to be modelled better with an increasing number of normal distributions.

- What values might you assume for the class prior probabilities if you had no estimate of their actual values?

We would assume they are all the same; no other reasonable choices are possible given that they have to sum to unity.

- Can you derive Bayes' theorem? This requires a consideration of joint probabilities and their properties.

The probability of two events occurring together is written as the joint probability $p(x, y)$. Since the events occur together there is no sense of one depending on the other or the "order" in which they occur. Thus $p(x, y) = p(y, x)$. A joint event can be expressed as

$$p(x, y) = p(x|y)p(y)$$

i.e. as the product of x being conditional on y occurring, multiplied by the probability that y will occur. Since the joint probability does not depend on the order of its arguments, we can write

$$p(x, y) = p(x|y)p(y) = p(y, x) = p(y|x)p(x)$$

so that

$$p(x|y)p(y) = p(y|x)p(x)$$

or

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

which is Bayes theorem.

Lecture 3. The maximum likelihood classifier: discriminant function and example

- Show that, under the assumption of equal class prior probabilities, the discriminant function for the Gaussian maximum likelihood classifier can be written as $g_i(\mathbf{x}) = -\ln|\mathbf{C}_i| - (\mathbf{x} - \mathbf{m}_i)^T \mathbf{C}_i^{-1} (\mathbf{x} - \mathbf{m}_i)$

The full discriminant function for the Gaussian maximum likelihood classifier is

$$g_i(\mathbf{x}) = \ln p(\omega_i) - \frac{1}{2} \ln |\mathbf{C}_i| - \frac{1}{2} (\mathbf{x} - \mathbf{m}_i)^T \mathbf{C}_i^{-1} (\mathbf{x} - \mathbf{m}_i)$$

If the priors are all equal, i.e. $p(\omega_i) = 1/N$ where N is the number of classes then the first term is the same for all $g_i(\mathbf{x})$ and can thus be removed because it doesn't add any discriminating information.

- If $\mathbf{C}_i = \mathbf{I}$ for all i what does the discriminant function then look like?

After removing terms which then don't aid discrimination among the classes, we are left with

$$g_i(\mathbf{x}) = \ln p(\omega_i) - \frac{1}{2}(\mathbf{x} - \mathbf{m}_i)^T(\mathbf{x} - \mathbf{m}_i)$$

which is the equation of a linear surface (hyperplane), noting that the inverse of the identity matrix is also the identity matrix..

- Generate the correlation matrices for the vegetation and fire burn classes for the classification exercise used in this lecture. Any comments? What about the water class?

The covariance matrices and the corresponding correlation matrices are

$$\mathbf{C}_V = \begin{bmatrix} 5.56 & 3.91 & 2.04 & 1.43 \\ 3.91 & 7.46 & 1.96 & 0.56 \\ 2.04 & 1.96 & 19.75 & 19.71 \\ 1.43 & 0.56 & 19.71 & 29.27 \end{bmatrix} \quad \mathbf{R}_V = \begin{bmatrix} 1.00 & 0.61 & 0.19 & 0.11 \\ 0.61 & 1.00 & 0.16 & 0.04 \\ 0.19 & 0.16 & 1.00 & 0.82 \\ 0.11 & 0.04 & 0.82 & 1.00 \end{bmatrix}$$

$$\mathbf{C}_{FB} = \begin{bmatrix} 9.38 & 10.51 & 12.30 & 11.00 \\ 10.51 & 20.29 & 22.10 & 20.62 \\ 12.30 & 22.10 & 32.68 & 27.78 \\ 11.00 & 20.62 & 27.78 & 30.23 \end{bmatrix} \quad \mathbf{R}_{FB} = \begin{bmatrix} 1.00 & 0.76 & 0.70 & 0.65 \\ 0.76 & 1.00 & 0.86 & 0.83 \\ 0.70 & 0.86 & 1.00 & 0.88 \\ 0.65 & 0.83 & 0.88 & 1.00 \end{bmatrix}$$

$$\mathbf{C}_w = \begin{bmatrix} 44.27 & 9.55 & 4.49 & 1.19 \\ 9.55 & 10.51 & 3.71 & 1.11 \\ 4.49 & 3.71 & 6.95 & 4.05 \\ 1.19 & 1.11 & 4.05 & 7.65 \end{bmatrix} \quad \mathbf{R}_w = \begin{bmatrix} 1.00 & 0.44 & 0.26 & 0.06 \\ 0.44 & 1.00 & 0.43 & 0.12 \\ 0.26 & 0.43 & 1.00 & 0.56 \\ 0.06 & 0.12 & 0.56 & 1.00 \end{bmatrix}$$

Remember they are addressed by row and column in the order: visible green, visible red, IR1 and IR2.

We note that the fire burn is a strongly correlated class overall whereas for vegetation, the correlations are, in general weaker, especially for the visible red and the second IR bands. Fire burn is highly correlated in that band pair. Water also shows weaker correlations overall. If we plotted a two-dimensional spectral space, with red and the second near IR bands as axes, the vegetation and water classes would look roughly circular whereas the fire burn class would look elongated, approximately along the diagonal of the space.

- The plots of the mean vectors for each of the four classes, shown opposite the thematic map, seems to suggest that the classes could easily be separated using just the IR1 band alone. Why is that not the case?

Those are plots of means. Looking at the covariance matrices we see that the vegetation and fire burn classes have large variances in the infrared bands, suggesting that there would be overlap if that band were used alone. Using the set of bands provides much better separation.

- For the case of equal class prior probabilities and two classes having a unit covariance matrix, show that the decision surface is a multidimensional linear surface.

The full discriminant function is

$$g_i(\mathbf{x}) = \ln p(\omega_i) - \frac{1}{2} \ln |\mathbf{C}_i| - \frac{1}{2} (\mathbf{x} - \mathbf{m}_i)^T \mathbf{C}_i^{-1} (\mathbf{x} - \mathbf{m}_i)$$

If the covariance matrix is the unity matrix the second term drops out as non-discriminating and the inverse covariance matrix in the third term has no effect. Thus the discriminant function becomes a linear expression. The decision surface between two classes is the difference between two linear functions, which itself is linear.

Lecture 4. The minimum distance classifier, background material

- Demonstrate that the two expressions below are the same

$$\mathbf{w}^T \mathbf{x} + w_{N+1} = 0 \quad \text{and} \quad \mathbf{w} \cdot \mathbf{x} + w_{N+1} = 0$$

This question asks you to show that $\mathbf{w}^T \mathbf{x}$ and $\mathbf{w} \cdot \mathbf{x}$ are equivalent.

For illustration let $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. The dot product $\mathbf{w} \cdot \mathbf{x}$ is formed by multiplying the vector elements in each row—e.g. $w_1 x_1$ —and summing the results.

The same outcome is obtained from the transpose expression:

$$\text{Note } \mathbf{w}^T = [w_1 \quad w_2 \quad w_3] \text{ so that } \mathbf{w}^T \mathbf{x} = [w_1 \quad w_2 \quad w_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

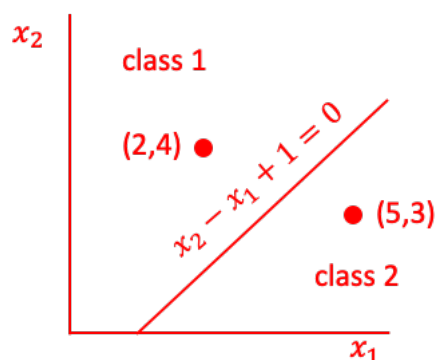
which is $w_1 x_1 + w_2 x_2 + w_3 x_3$

Using a two-dimensional example verify the decision rule

$$\mathbf{x} \in \text{class 1 if } \mathbf{w}^T \mathbf{x} + w_{N+1} > 0$$

$$\mathbf{x} \in \text{class 2 if } \mathbf{w}^T \mathbf{x} + w_{N+1} < 0$$

Using the diagram below we want to test the value of the polynomial $x_2 - x_1 + 1$ for the two points indicated.



For (2,4) $x_2 - x_1 + 1 = 3$. As expected, this is positive for class 1.

For (5,3) $x_2 - x_1 + 1 = -1$. As expected, this is negative for class 2.

Lecture 5. Training a linear classifier

- Demonstrate that $\mathbf{y}^T \mathbf{z} = \mathbf{z}^T \mathbf{y}$

If $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ and $\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$ then $\mathbf{y}^T \mathbf{z} = [y_1 \ y_2 \ y_3] \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$ and $\mathbf{z}^T \mathbf{y} = [z_1 \ z_2 \ z_3] \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$

which, when multiplied out are both the same. This can also be done by noting that $\mathbf{y}^T \mathbf{z} = \mathbf{y} \cdot \mathbf{z}$ and the fact that the dot product is commutative.

- What is the slope of the decision surface in the minimum distance rule for a two class, two-dimensional example?

The general equation for the decision surface is

$$2(\mathbf{m}_j - \mathbf{m}_i)^T \mathbf{x} + (\mathbf{m}_i^T \mathbf{m}_i - \mathbf{m}_j^T \mathbf{m}_j) = 0$$

$2(\mathbf{m}_j - \mathbf{m}_i)^T$ is a column vector, with two elements in a two-dimensional space.

Call it $\begin{bmatrix} m_1 \\ m_2 \end{bmatrix}$. Likewise, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. The last part of the equation above will be a scalar.

Thus the slope of the decision surface will be

$$\frac{dx_2}{dx_1} = -m_1/m_2$$

- What do you think would be the optimum location for the decision surface in the two class data set considered in this lecture?

The minimum distance classifier places it mid-way between class means, but a better position might be in the middle of the gap between the two classes. We will look at that in detail when we come to the support vector classifier.

- Suppose you have a method for finding the optimal decision surface between two classes. How might you construct a multi-class algorithm?

We could set up a decision tree of binary decisions. In training we could first separate one class from the rest. Then we could separate out another class from the remainder, and so on until at the end we are separating just the last two classes. There are more efficient decision tree methods, again which we will meet in the context of the support vector classifier.

Lecture 6. The support vector machine: training

- Demonstrate that $w_1x_1 + w_2x_2 + \dots w_Nx_N + w_{N+1} = 0$ is the same as $\mathbf{w}^T \mathbf{x} + w_{N+1} = 0$

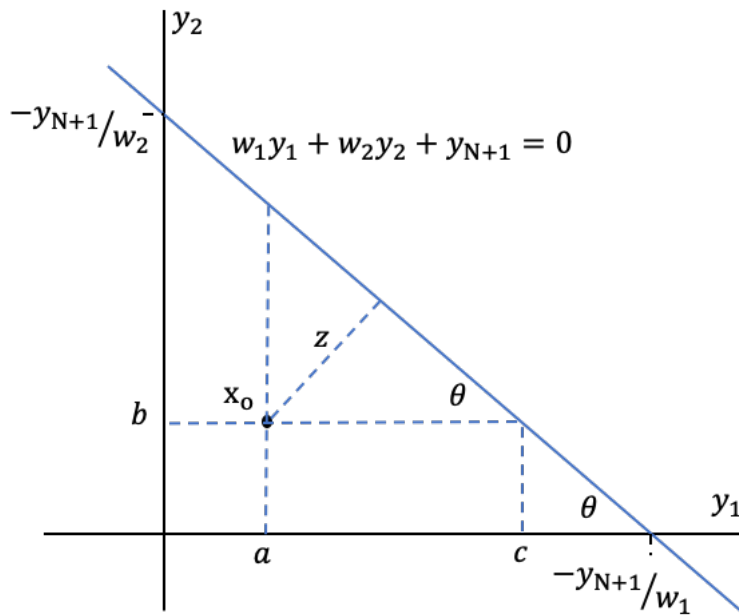
This makes use of $\mathbf{w}^T \mathbf{x} = \begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = w_1x_1 + w_2x_2 + \dots w_Nx_N$

- Can you derive $\frac{|\mathbf{w}^T \mathbf{x}_0 + w_{N+1}|}{\|\mathbf{w}\|}$ for the simple case of two dimensions?

We will use the following simple diagram to do the derivation, which will involve finding the sine of the angle θ from two perspectives and equating the results.

From the large triangle formed by the line and the axes we have

$$\begin{aligned} \sin(\theta) &= \frac{-y_{N+1}/w_2}{\sqrt{(-y_{N+1}/w_2)^2 + (-y_{N+1}/w_1)^2}} \\ &= \frac{1/w_2}{\sqrt{(1/w_2)^2 + (1/w_1)^2}} \\ &= \frac{w_1}{\sqrt{w_1^2 + w_2^2}} = \frac{w_1}{\|\mathbf{w}\|} \end{aligned}$$



Now from the smaller triangle we can see that

$$\sin(\theta) = \frac{z}{c - a}$$

where $c = -(w_2 b + y_{N+1})/w_1$ so that

$$\sin(\theta) = \frac{zw_1}{-[w_1 a + w_2 b + y_{N+1}]}$$

Equating the two expressions for the sine, we have

$$\frac{zw_1}{-[w_1 a + w_2 b + y_{N+1}]} = \frac{w_1}{\|\mathbf{w}\|}$$

so that

$$z = \frac{|-[w_1 a + w_2 b + y_{N+1}]|}{\|\mathbf{w}\|} = \frac{|\mathbf{w}^T \mathbf{x}_0 + y_{N+1}|}{\|\mathbf{w}\|}$$

where $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ and $\mathbf{x}_0 = \begin{bmatrix} a \\ b \end{bmatrix}$.

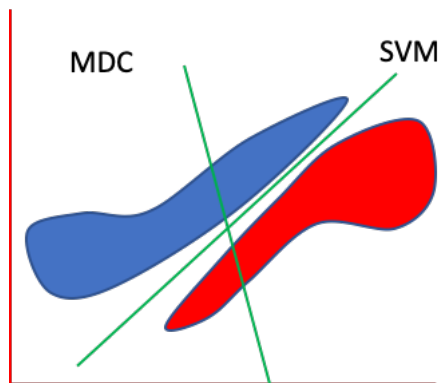
- Verify $\|\mathbf{w}\|^2 = \mathbf{w}^T \mathbf{w}$

$$\mathbf{w}^T \mathbf{w} = [w_1 \quad w_2 \quad w_3 \quad \dots] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \end{bmatrix} \text{ which is the sum of the squares of the elements,}$$

and thus the square of the norm (or size).

- Can you demonstrate how a support vector classifier is likely to perform better than the minimum distance classifier by choosing an unusual distribution of training pixels?

In the following diagram the means of the two classes will be strongly influenced by the major concentrations of pixels as indicated, leading to an unacceptable decision surface. On the other hand, the SVM is only influenced by the pixels near the margins and is unaffected by the unusual concentrations on pixels at either ends of the distributions.



Lecture 7. The support vector machine: the classification step and overlapping data

- Why does $\sum_i \xi_i$ represent a good measure of the pixels in error, during training?
The slack variables ξ_i are all positive; they are zero only for those pixels that are on the corrects sides of the separating hyperplane. Therefore, their sum is an indication of the pixels that are not optimally or correctly located with respect to the separating hyperplane.
- Why is $\mathbf{w}$ a vector in the expression $\mathbf{w} = \sum_{i \in \mathcal{S}} \alpha_i y_i \mathbf{x}_i$?
Because the α_i and y_i are scalars and the sum is taken over the set of support vectors.
- How might you choose a value for the regularization parameter C ?
Without any prior knowledge one approach would be to choose a representative set of training data and then run a number of trials using different values of C to see

which is best. That might be done through some form of searching, moving from coarse differences in C to finer differences as the optimal value is approached.

Lecture 8. The support vector machine: non-linear data

- The radial basis function kernel $k(\mathbf{x}_i, \mathbf{x}) = \exp\{-\gamma\|\mathbf{x} - \mathbf{x}_i\|^2\}$ has a parameter γ . A value needs to be found for γ before the kernel can be used. Can you suggest how that might be done in practice?

As in the answer to the last question concerning a value for C , an acceptable value for γ would be found by running a number of trials, and choosing the value of γ that leads to the best performance.

- Is $\mathbf{x}_i^T \mathbf{x}$ a valid kernel?
Yes, because it is a simple scalar product.
- There are a set of rules that can be used for generating kernels. Two are:

$$k(\mathbf{x}_i, \mathbf{x}) = \exp\{k_1(\mathbf{x}_i, \mathbf{x})\}$$

$$k(\mathbf{x}_i, \mathbf{x}) = f(\mathbf{x}_i)k_1(\mathbf{x}_i, \mathbf{x})f(\mathbf{x}) \text{ where } f(\mathbf{x}) \text{ is any function}$$

$$\text{Noting that } \|\mathbf{x} - \mathbf{x}_i\|^2 = \mathbf{x}^T \mathbf{x} - 2\mathbf{x}^T \mathbf{x}_i + \mathbf{x}_i^T \mathbf{x}_i$$

show that the radial basis function above is a valid kernel

The radial basis function can be expanded as

$$k(\mathbf{x}_i, \mathbf{x}) = \exp\{-\gamma\|\mathbf{x} - \mathbf{x}_i\|^2\} = \exp\{-\gamma\mathbf{x}^T \mathbf{x}\} \exp\{2\gamma\mathbf{x}^T \mathbf{x}_i\} \exp\{-\gamma\mathbf{x}_i^T \mathbf{x}_i\}$$

Since the exponential function of a valid kernel is itself a valid kernel then the middle exponential term in this last expression is valid. The outer exponential functions occur in the form of the second generating rule above, so that overall the radial basis function is valid as a kernel.

Lecture 9. The support vector machine: multiple classes and the classification step

- A real thematic mapping exercise in remote sensing will involve more than two classes. To cope with that the binary SVM classifier is used, as we have seen, in a decision tree framework. Suppose we adopt a one-against-one strategy for an exercise involving 16 classes.
 - How many separate SVMs are needed?
 $M(M-1)/2$ separate classifiers are needed, which for $M=16$ is 120.

- If a radial basis function kernel is chosen for all of the SVMs how many unknowns (kernel and regularization parameters) have to be found from training data?
Usually the same parameters are used for all the SVMs, so the answer is 2.
- Is it practical to find values for all of the individual parameter sets?
If separate values of the parameters were to be found for each of the classifiers, there would be a total of 240. That is a very large number to find while, at the same time, trying to identify the support vectors. Training would be complex and time-consuming and there may not be enough training samples available to estimate that number of unknown reliably.
- What would be a preferred practical approach?
As above, to use the same values of the regularisation and kernel parameters for all SVMs in the decision tree. Experience has shown that classifier performance is not strongly dependent on getting exact values for those parameters so that using the same values over all classifiers is usually acceptable

Lecture 10. The support vector machine: an example

- Why is the SVM preferable to the maximum likelihood classifier for hyperspectral image data?
There are no multidimensional second order statistics to be estimated with the SVM, whereas the covariance matrix has to be estimated for the MLC. As dimensionality increases, say with hyperspectral data sets, the number of training samples needed rises beyond levels that are practical for use with the MLC; classification errors, usually described by the term Hughes phenomenon, occur.
- Could the minimum distance classifier be used successfully on hyperspectral data sets?
Yes, because the central operation in the separating hyperplane is a scalar product of two vectors—one involving the class means and the pixel vector. No second order statistics have to be estimated, only the class means.
- Outline the steps to be followed to find satisfactory values for the regularization and kernel parameters for an SVM. Is the set of support vectors affected?
A grid search process is used, in which a two-dimensional search space, with axes defined by the two parameters, is set up and searched for the best pair of values. That is usually done on a representative training set of data. Because the regularization parameter changes over the search, the set of support vectors will be affected.

Lecture 11. The neural network as a classifier

- Verify the values of the activation function $\frac{1}{1+\exp(-z/b)}$ for $z = 0, -\infty, +\infty$. What is its slope at the origin?

In the limit as $z \rightarrow \infty$, $\exp(-\frac{z}{b}) \rightarrow 0$ so that the activation function approaches 1

In the limit as $z \rightarrow -\infty$, $\exp(-\frac{z}{b}) \rightarrow \infty$ so that the activation function approaches 0

At $z = 0$, $\exp(-\frac{z}{b}) = 1$ so that the activation function is 0.5

If we represent the activation function by f , then $\frac{df}{dz} = \frac{1}{b} \exp(-\frac{z}{b}) f^2$ which, at $z=0$ is $\frac{1}{4b}$.

- How many weights will there be in a neural network with 6 input nodes, a single hidden layer of 18 nodes and an output layer with 12 nodes? Include the thresholds θ in the calculations.
 $(6+1) \times 18 + (18+1) \times 12 = 354$
- The squared difference error measure chosen to assess the performance of the neural network is a little different to the usual measures of error, such as root mean square, sum of the absolute differences, etc. Can you guess why the sum of squared measure is chosen for neural networks, and why it is an acceptable measure to use?
The RMS measure requires the manipulation (differentiation) of a square root operation in the analysis, while the sum of absolute differences is also difficult to manipulate when developing the training algorithm. By contrast the SSE is easily differentiated in the back-propagation training method.

Lecture 12. Training the neural network

- Can you foresee any problems with the use of the gradient descent method of weight adjustment, especially if the error curve is more complex than the simple parabola used for illustration in this lecture?
It does not guarantee that local, as against the global, minimum of the error curve will be found.
- Would it be preferable to use a large or small value of the learning parameter η ? What are the consequences of large and small choices?
It is hard to say in general which would be preferred but we should note that a large value may over-adjust, and thus generate new weight values that are on "the other side" of the minimum in the error curve. The next adjustment might then cause it to go back to the first side, thus leading to oscillations (or instability) in the overall adjustment process. A smaller value will avoid that problem by moving carefully down the error curve but, if too small, slows convergence.

- Why would scaling the inputs sometimes be desirable when designing the input layer?
If the different features in the measurement vector fed in to the NN have different dynamic ranges—which could happen if the single input vector is constructed by concatenating the measurements from different sensors—then there may be benefit if all features lay in the same range of values, so that some don't dominate the learning process. That can be done by scaling at the input layer.
- What would be the consequence of choosing identical starting values for the weights?
The network cannot be trained, since unequal weights cannot then be found. See D.E. Rumelhart, G.E. Hinton and R.J. Williams, Learning internal representations by error propagation, in D.E. Rumelhart and J.L. McClelland (eds), *Parallel Distributed Processing: Explorations in the Microstructures of Cognition*, vol 1: Foundations, pp 318-362, MIT Press, Cambridge, Mass.

Lecture 13. Neural network examples

- Verify the output layer values shown in the fourth slide of this lecture
From the second slide, the output equation is $g_3 = f(z_3)$, $z_3 = w_5 g_1 + w_6 g_2 + \theta$ where f is the sigmoid activation function: $\frac{1}{1+\exp(-z/b)}$

The final values for the weights and offset are $w_5 = 4.785$ and $w_6 = 4.784$, $\theta = -2.493$ and while we chose $b = 1$ in the initial specification of the network.

Thus for $g_1 = 0$ and $g_2 = 0$ we have $z_3 = -2.493$, giving $g_3 = 0.076$
Thus for $g_1 = 0$ and $g_2 = 1$ we have $z_3 = 2.291$, giving $g_3 = 0.908$
Thus for $g_1 = 1$ and $g_2 = 0$ we have $z_3 = 2.292$, giving $g_3 = 0.908$
Thus for $g_1 = 1$ and $g_2 = 1$ we have $z_3 = 7.076$, giving $g_3 = 0.999$
- What properties should the activation function desirably have?
It should be able to distinguish between two categories when a threshold is applied, and it must be differentiable so that the back-propagation training process can be used to learn the network parameters.
- Can you think of any suitable activation functions other than the one used in this example?
Any function which has the properties above would be suitable. The hyperbolic tangent $\tanh$ is an example.
- By inspecting the error versus iteration curves for the two examples given in this lecture justify why a momentum term should be used in the weight updating formula.

The error curves in both examples are smooth, downwards monotonic functions, which suggests that training would be benefited and not disadvantaged by the incorporation of a momentum term in the updating formula.

Lecture 14. Deep learning and the convolutional neural network, part 1

- Comment on the importance of the spatial context of pixel in the following circumstances:
 - Low spatial resolution data (say a pixel size of 1km) over mixed agriculture and horticulture in a small country, such as New Zealand
Less important because of the ratio of spatial resolution and likely field size
 - High spatial resolution data (say a pixel size of 10m) over broad acre agriculture in a large country, such as Australia
Important because of the ratio of spatial resolution and likely field size
 - Medium spatial resolution data (say a pixel size of 30m) over an urban region
Less important because of the ratio of spatial resolution and likely field size
 - Very high spatial resolution data (say a pixel size on 0.1m) over an urban region
Important because of the ratio of spatial resolution and likely field size
- In an ordinary (fully connected) neural network do the number of weights between the input and the first hidden layer depend upon the number of pixels, if presented with a full image?
Yes, because there is a set of input nodes associated with each pixel (one for each element of the pixel vector) and this is replicated for each pixel. All those input nodes are then connected to all nodes in the first hidden layer.
- In a CNN do the number of weights between the input layer and the first hidden layer depend upon the number of pixels, if presented with a full image?
Yes, but because of the much smaller local receptive field and weight re-use, the number is considerably smaller than in the previous question.
- Why should multiple hidden layers be beneficial?
Each layer implements sets of linear decisions; when cascaded in multiple layers, quite complicated piecewise linear decisions can be made. A good example of this is seen in Fig 8.5 of Y-H Pao, *Adaptive Pattern Networks and Neural Networks*, Addison-Wesley, NY., 1989.
- What is the disadvantage of having multiple hidden layers?
A large increase in the number of connections and thus weights and offsets, all of which have to be learned through training.

Lecture 15. Deep learning and the convolutional neural network, part 2

- Consider a 100x100 pixel single-band image. Suppose the receptive field in a CNN is 5x5. What will be the size of the first convolution layer, without zero padding, and with a stride of 1?

Without zero padding there will be a loss of two columns on both sides of the image and two rows at the top and the bottom, leaving a convolutional layer of 96x96 nodes.

- Can you devise a formula for the size of the first convolution layer of the previous question?

Suppose the dimension of a square image is $N \times N$ pixels, the receptive field (or filter) size is $F \times F$, and the first convolutional layer $C \times C$. The value of C is the dimension of the image less the rows and columns lost on each edge because of the need to centre the filter so that a full receptive field is seen. That loss will be $(F - 1)/2$ per side, giving $(F - 1)$ in each dimension. Thus $C = N - (F - 1)$. For the above example that is $100 - (5 - 1) = 96$.

- What will be the size of the subsequent pooling layer if pixels are pooled in 2x2 groups?

Assuming that the pooling is done with a stride of 2, which is common for 2x2 pooling, the pooling layer will have halve the number of rows and columns—for this example that is 48.

- Demonstrate that the softmax "probabilities" sum to unity and lie in the range (0,1)

The softmax probability is defined by $p(o_n = t_n) = \frac{e^{o_n}}{\sum_{n=1}^N e^{o_n}}$. If we sum this formula over all n , then the result is unity. If there is only one event (which thus has a probability of unity) then $N = 1$ in the formula giving $p(o_n = t_n) = 1$ as required. It can never be zero (from the formula) but can be very close. For example if the two outputs of a binary classification were 1 and 100, the two softmax probabilities are 0.0001 and 0.9999.

- Is the ReLU function differentiable?

Yes, except at the origin, which is a discontinuity. For negative arguments the derivative is zero, while for positive arguments it is the slope of the line—unity.

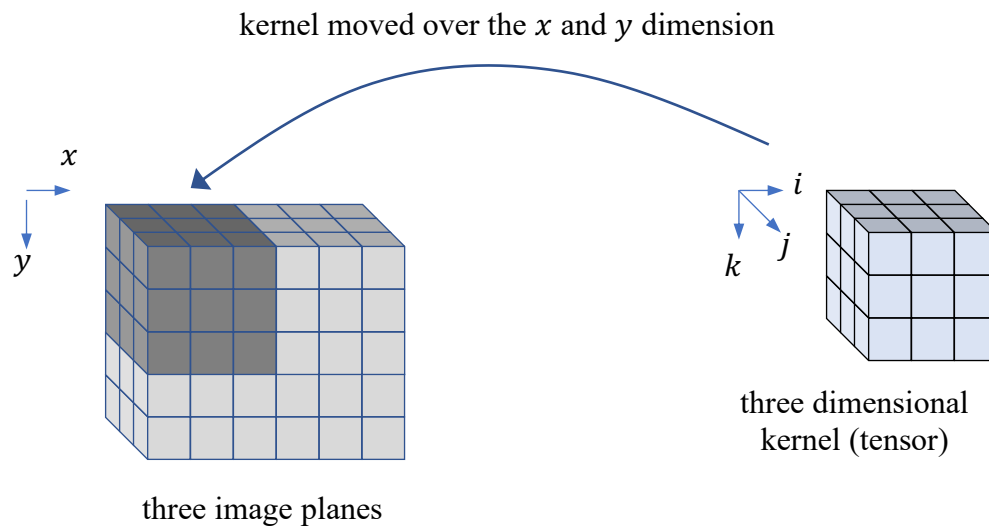
Lecture 16. Deep learning and the convolutional neural network, part 3

- If you wished to reduce the dimensionality of hyperspectral data by using a principal components transform prior to inputting it into a CNN what measure might you use to determine how many principal components to retain?

We would determine how much image variance it was important to retain and then select the number of PCs, starting from the first, which account for that amount.

Recall that the variance retained is the sum of the corresponding eigenvalues as a fraction of the sum of all eigenvalues.

- The diagram below shows a three dimensional convolution kernel. Describe its operation mathematically. Is it any different from applying three separate two dimensional kernels and summing the results?



Let the image pixel values be represented by $I(c, i, j)$ addressed according to the elements of the kernel—in other words, the indices i, j take the place of x, y when the filter is placed at the position x, y in the image itself. The letter c is an index for the colour or spectral dimension of the image. Suppose it has values 1,2,3. Let the corresponding kernel (filter) entries be $w(c, i, j)$. Then one step of the three-dimensional filtering operation can be described by

$$R(x, y) = \sum_{c, i, j} w(c, i, j) I(c, i, j)$$

which can be expanded as

$$R(x, y) = \sum_{i, j} w(1, i, j) I(1, i, j) + \sum_{i, j} w(2, i, j) I(2, i, j) + \sum_{i, j} w(3, i, j) I(3, i, j)$$

which is the same as applying the filters separately, as in the lectures. The offset would be added to this sum before the activation function is applied in a CNN.

Lecture 17. CNN examples in remote sensing

- List the set of unknowns that generally have to be evaluated in training a CNN
If the ReLU is used there is no need to specify any activation function parameter. The learning rate parameter, and momentum parameter if that is used, have to be specified. Then the full set of weights and offsets needs to be found through training.

- Verify that 81,408 unknowns need to be found in the first example treated in this lecture

There are 24 weights plus 1 offset between the input and the convolutional layer for each filter, giving a total of 500.

There are 5 connections between the convolution and pooling layers for each filter path, but they don't add any unknowns.

There are $20 \times 40 = 800$ inputs to the first fully connected (hidden) layer. That layer has 100 nodes, so the number of weights and offsets connecting the input and hidden layers is $(800+1) \times 100 = 80,100$.

The number of weights and offsets connecting the hidden layer to the output layer is $(100+1) \times 8 = 808$.

Altogether that gives $500 + 80,100 + 808 = 81,408$

- Are there enough training samples in the above to ensure good parameter estimation?

Yes, as noted in the lecture 1600 training pixels are available. Each pixel has 220 features (bands) giving a total of $1600 \times 220 = 352,000$ training samples. That averages at just over 4 samples per unknown, which should give sufficiently reliable estimates.

- How many spectral plus spatial features are fed to the first fully connected layer in the second example treated in this lecture?

Assume for simplicity that padding is used with the convolution operations so that there is no loss of dimensionality when using the convolution filters.

In the spectral branch the data vector has 200 channels; with 5×1 pooling that is reduced to 40. There are 16 spectral channels, so that the number of spectral features to be fed into the fully connected layer is 640.

The patch size used in the spatial branch is 21×21 ; after 2×2 pooling this reduces to 10×10 (ignoring the small difference). With 30 filters that means there would be 3000 spatial features to be fed into the fully connected layer, giving a total (spatial plus spectral) of 3640.

Lecture 18. Comparing the classifiers

- Which of MLC, SVM and CNN would you use as a classifier in each of the following situations?

- Mapping crops fields using Landsat ETM data on a weekly basis throughout the growing season

This simple exercise could use MLC

- Detecting diseased trees in a natural forest using hyperspectral aircraft imagery

Because the data is hyperspectral either SVM or CNN might be used. SVM would be simpler.

- Detecting objects in high spatial resolution imagery
Object detection implies recognizing spatially connected features. The CNN would be used.
- Producing a map of cover types in an urban landscape using high spatial resolution RGB imagery
Since the spectral dimensionality is low, MLC might be the simplest to use.

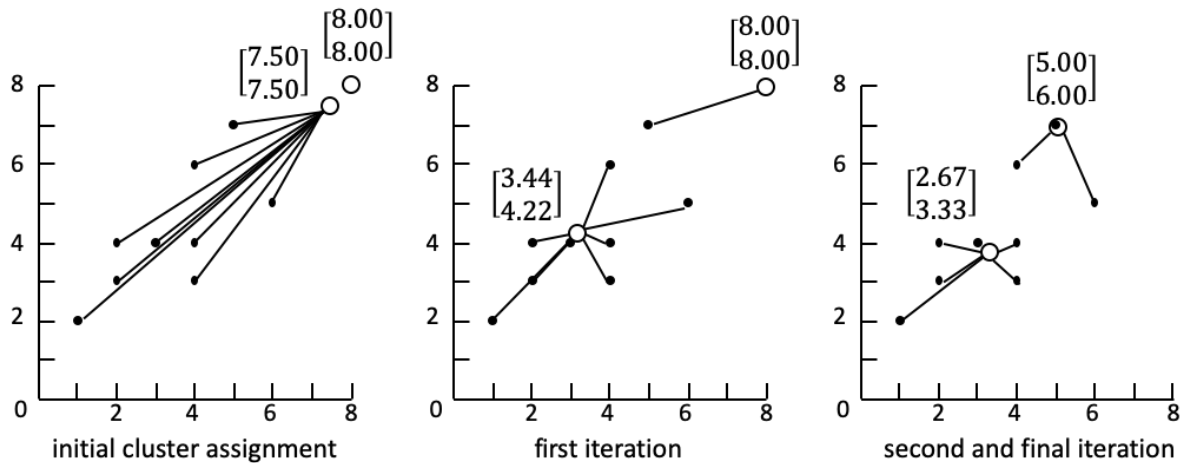
Lecture 19. Unsupervised classification and clustering

- When using the moving means clustering algorithm
 - Do the initial cluster means need to be all different?
Yes, otherwise multiple clusters will not be generated
 - How might you estimate how many clusters to look for?
The rule of thumb is about 2 to 3 times the number of identifiable ground cover classes.
 - How might you choose a good initial distribution of cluster means?
Ideally, they should be well-spaced (uniformly?) over the spread of the pixel vectors in spectral space. A convenient approximation to that is to choose initial cluster centres that are spaced along the spectral diagonal of the data.
 - Why is it impractical to attempt to examine every possible clustering of a set of N pixels into C clusters?
There are $C^K / K!$ ways of placing K pixels into C clusters, which is a very large number of possibilities to check for practical values of C and K .
 - Do you think that the result of a moving means clustering is unique?
No, it depends upon the number of initial clusters selected and their starting positions.

Lecture 20. Examples of k means clustering

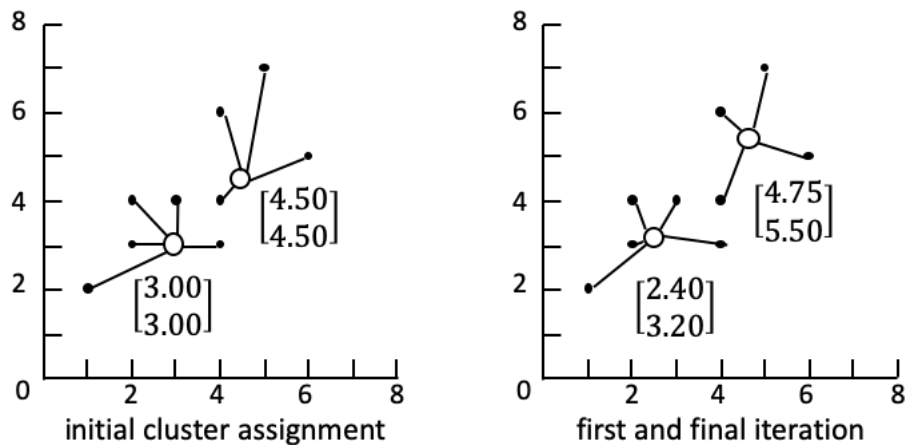
- In the first example of this lecture, comment on what might happen if you chose the initial cluster centres as
 - (7.5, 7.5) and (8.0, 8.0)
All points are closer to (7.5, 7.5) than (8.0, 8.0) as a result of which no pixels would be assigned to the second cluster initially. The mean of the first cluster is then the global mean, which is $\begin{bmatrix} 3.44 \\ 4.22 \end{bmatrix}$. The top point is now closer to (8.0, 8.0) and migrates

to that cluster on the next iteration, starting a general movement of pixels to clusters as seen below. After the second iteration there is no further movement of pixels between clusters, so that it is the final state. This is a slightly different clustering than the example in lectures.



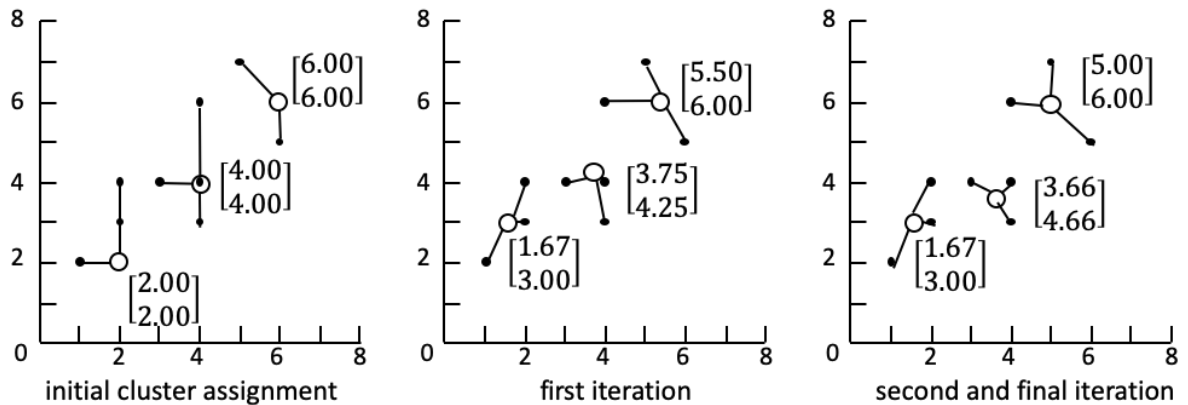
- (3.0, 3.0) and (4.5, 4.5)

In this case, because the initial cluster means are so close to their final values, only one iteration is needed to find a stable configuration. Note that it is the same result as in the example in the lectures.



- (2.0, 2.0), (4.0, 4.0) and (6.0, 6.0)

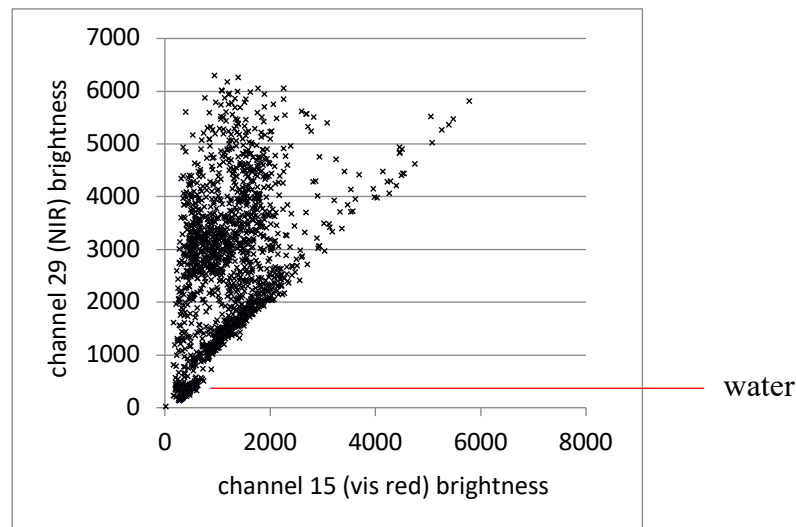
This is interesting because two of the pixels could have been allocated to one of two of the clusters. In the following they are put in the cluster encountered first in a sequential process. Also, the three clusters are probably different from a choice made by inspecting the original data set.



- Explain the shape of the “bare” spectral reflectance curve constructed from the cluster means in the second example in this lecture.
It rises monotonically with wavelength in the visible and near IR wavelengths, indicative of a surface with little vegetation, and then falls in the mid IR, again consistent with soil with some degree of moisture content.
- In those same curves explain why those for “trees” and “sparse veg” don’t show lower values in the visible red bands.
Because neither has strong chlorophyll content, and in both cases there would be a significant component of background—presumably soil. Thus, the amount of red absorption is not as great as it would be for healthy green vegetation along with the fact that the observed responses are mixtures of trees/sparse veg and soil responses.

Lecture 21. Other clustering methods

- How important is the first line of samples to the operation of the single pass algorithm?
It controls the generation of initial cluster centres and thus influences how clusters eventually evolve.
- Suppose you had an image that consisted of a body of water at the top of the image. What does that imply for single pass clustering?
In principle it would only initiate a single cluster.
- Really, what is a cluster? What clusters can you actually see in the following scatter plot? Doesn’t a lot of the data look like it forms part of a continuum?
We talk about clusters as distinct groups of pixels in spectral space. Sometimes that is correct, as in the water class below, but often the data space is a continuum or a distribution with no discernable groups, as in the vegetated and soil regions below. However, we still use clustering algorithms to segment the space by grouping pixels together even though they don’t look like compact groups, as we think of as a cluster to be. We will have more to say about that under “classification methodologies” in module 3.



Lecture 22. Clustering “big data”

- How important is it to use all pixels when undertaking clustering as method for supervised classification?

There are several aspects to this question. First, when using clustering for unsupervised classification, we need to find the cluster centres and then allocate all the pixels of the image to one of the clusters using a minimum distance assignment to the closet cluster mean. Clearly, all pixels are involved in this last step, but only a representative sample is needed to generate the clusters and their mean vectors. So, for cluster generation, it is not important to use all the pixels in the image.

- If you answered “not important” to the previous question how would you select samples to cluster that are representative of the scene?

Obviously, the samples need to be selected so that they are representative of all the apparent classes in the image. Statistically, the best way would be to do random sampling or even random sampling stratified by apparent cover type. In practice though it is often more convenient to take (heterogeneous) regions of pixels that are representative of the image and cluster those sets of pixels. An important benefit of doing that is that boundary pixels between classes, and pixels that are mixtures of classes, are picked up as separate clusters.

- Do the k means and K trees methods generally require all the data to be clustered to be resident in the core memory of a computer, as against being held in disc memory? What about the single pass method we looked at earlier?

In general, yes, because they involve iteration over the full data set (k means) or subsets of the data (K trees). By contrast the single pass algorithm does not require all the data to be core memory resident for its operation.

- If you had undertaken clustering with the K trees method, based on a given set of pixel vectors, and then another representative pixel became available for training, how would insertion of that new pixel be handled by the tree; in other words, how would the tree be modified? What would be the case if you had used the k means approach instead?

A benefit of the K trees method is that new pixels can simply be inserted into the tree and they flow down to their correct leaf node. If tree capacity is exceeded at any stage, then the tree structure is incrementally adjusted. By contrast, if new pixels become available to the k means method the process, in principle, has to be re-started.